Analytical Geometry & Calculus1 (MTH111)

Nile University

Lecture #3

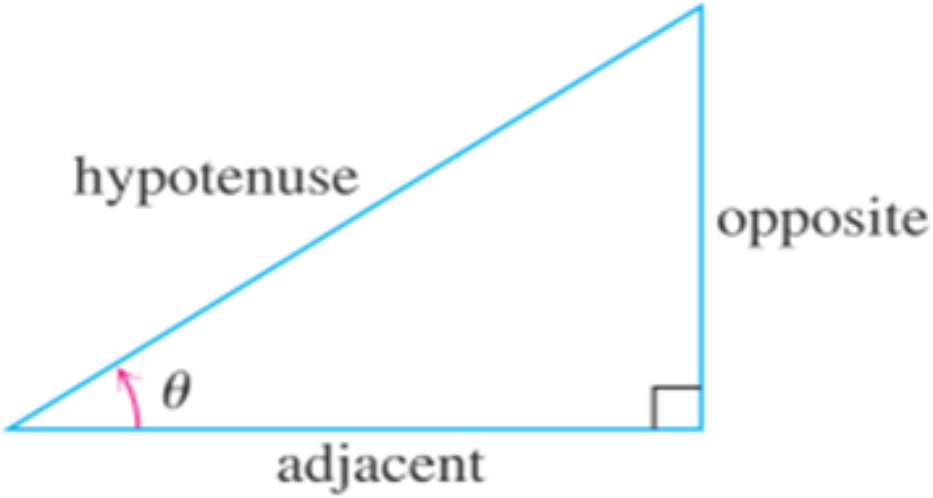
- Trigonometric Functions
- Inverse Trigonometric Functions
- Derivatives of Inverse Trigonometric Functions

Trigonometric Functions

2

$$\sin\theta = \frac{\text{opp}}{\text{hyp}}$$

$$\cos\theta = \frac{\text{adj}}{\text{hyp}}$$



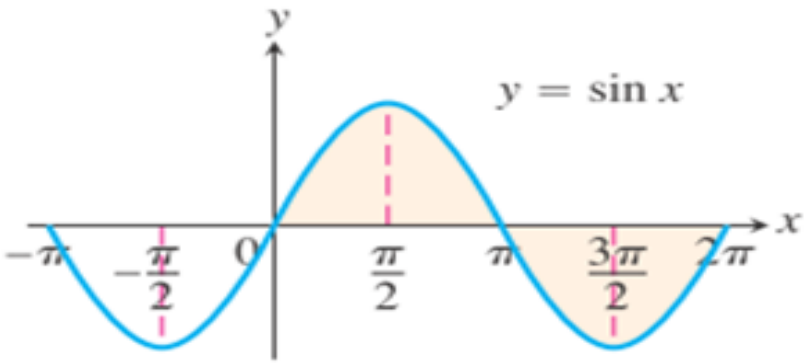
Note: $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$

$$\csc\theta = \frac{1}{\sin\theta}$$

$$\sec\theta = \frac{1}{\cos\theta}$$

$$\cot\theta = \frac{1}{\tan\theta} = \frac{\cos\theta}{\sin\theta}$$

Trigonometric Functions

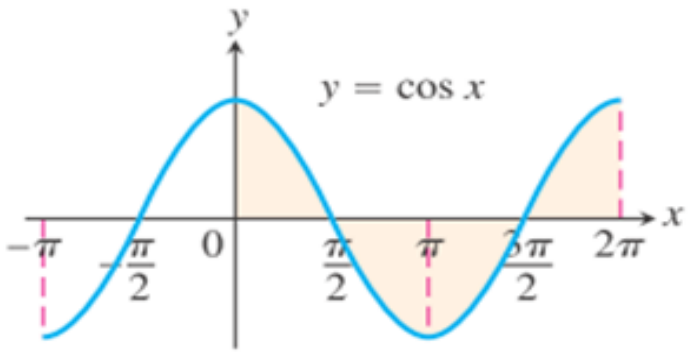


Domain: $-\infty < x < \infty$

Range: $-1 \le y \le 1$

Period: 2π

Odd function



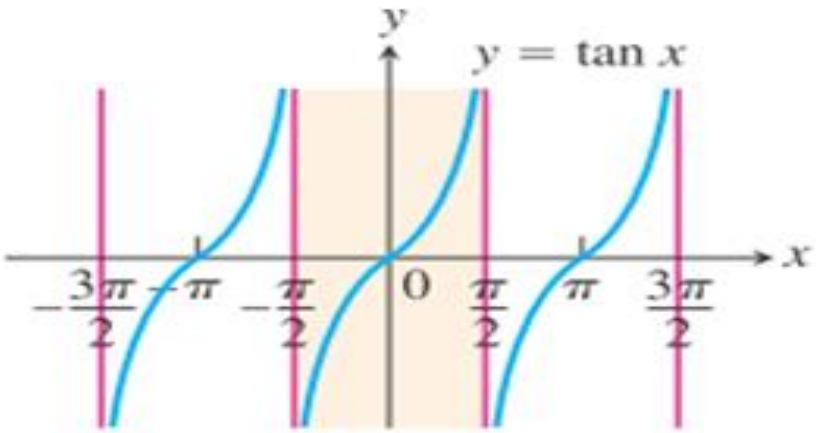
Domain: $-\infty < x < \infty$

Range: $-1 \le y \le 1$

Period: 2π

Even function

Trigonometric Functions



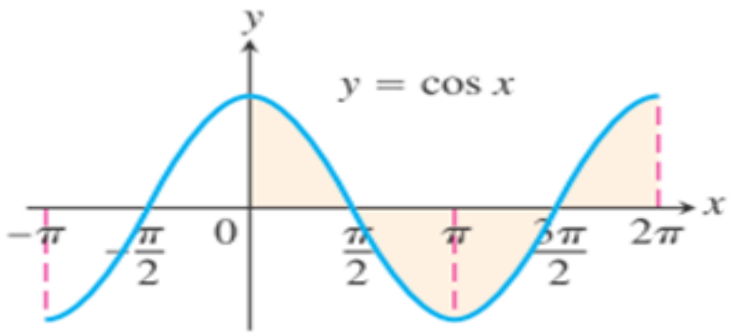
Domain: $x \neq \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \ldots$

Range: $-\infty < y < \infty$

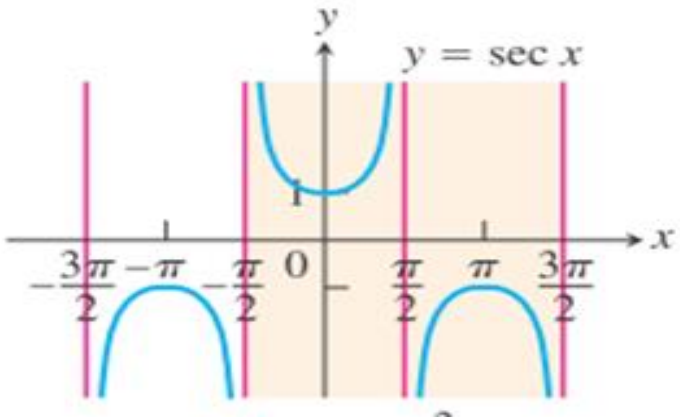
Period: π

Odd function

Trigonometric Functions



$$\sec\theta = \frac{1}{\cos\theta}$$ **Reciprocal**



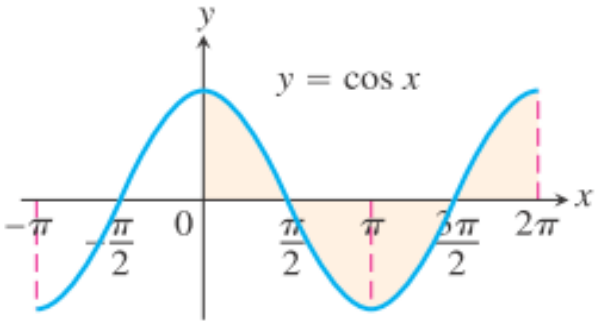
Domain: $x \neq \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \ldots$

Range: $y \leq -1$ or $y \geq 1$

Period: 2π

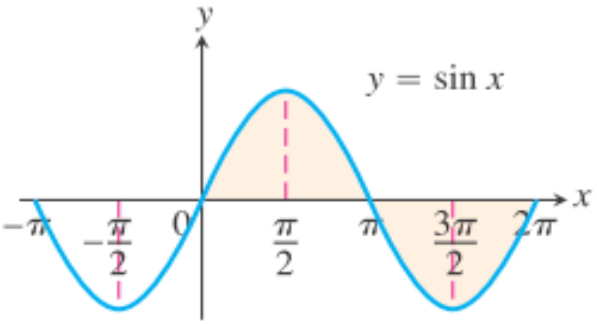
Even function

Trigonometric Functions(summary)



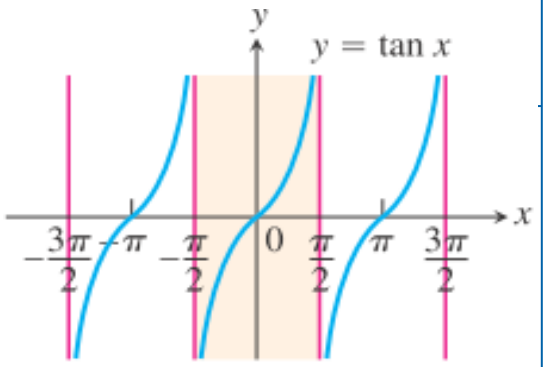
Domain: $-\infty < x < \infty$
Range: $-1 \le y \le 1$
Period: 2π

(a)



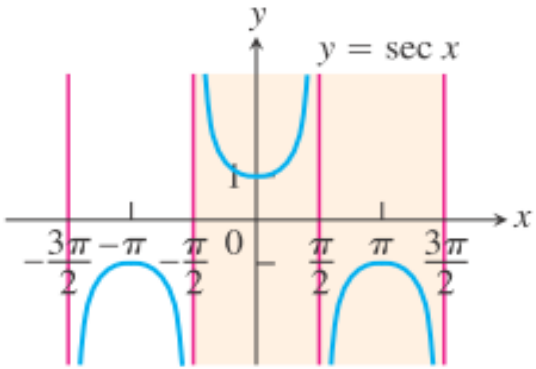
Domain: $-\infty < x < \infty$
Range: $-1 \le y \le 1$
Period: 2π

(b)



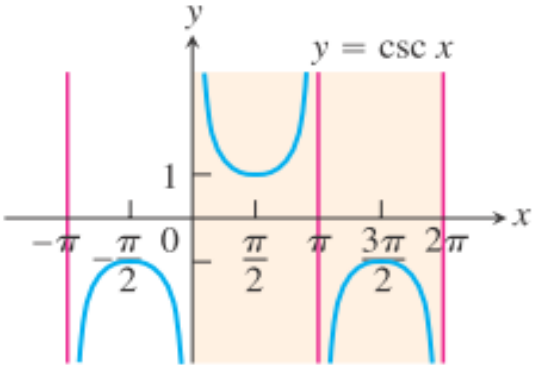
Domain: $x \neq \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \ldots$
Range: $-\infty < y < \infty$
Period: π

(c)



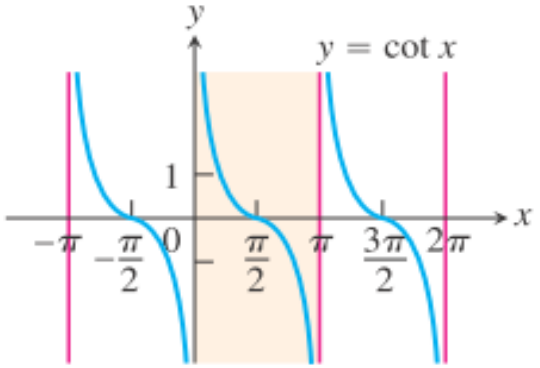
Domain: $x \neq \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \ldots$
Range: $y \le -1$ or $y \ge 1$
Period: 2π

(d)



Domain: $x \neq 0, \pm\pi, \pm 2\pi, \ldots$
Range: $y \le -1$ or $y \ge 1$
Period: 2π

(e)



Domain: $x \neq 0, \pm\pi, \pm 2\pi, \ldots$
Range: $-\infty < y < \infty$
Period: π

(f)

Trigonometric Identities

$$\cos^2\theta + \sin^2\theta = 1$$

$$1 + \tan^2\theta = \sec^2\theta$$

$$1 + \cot^2\theta = \csc^2\theta$$

Addition Formulas

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

Double-Angle Formulas

$$\cos 2\theta = \cos^2\theta - \sin^2\theta$$

$$\sin 2\theta = 2 \sin\theta \cos\theta$$

Half-Angle Formulas

$$\cos^2\theta = \frac{1 + \cos 2\theta}{2}$$

$$\sin^2\theta = \frac{1 - \cos 2\theta}{2}$$

Inverse Trigonometric Functions

$\sin^{-1}(x)$ $\cos^{-1}(x)$

$\tan^{-1}(x)$ $\cot^{-1}(x)$

$\sec^{-1}(x)$ $\csc^{-1}(x)$

Note that $\boldsymbol{\sin^{-1}(x) \neq \frac{1}{\sin(x)}}$

$\sin^{-1}(x)$ is the inverse of $\sin(x)$

$\sin(\frac{\pi}{2})=1$ $\sin^{-1}(1)=\frac{\pi}{2}$

$\sin(y)= x$ $\sin^{-1}(x)=y$

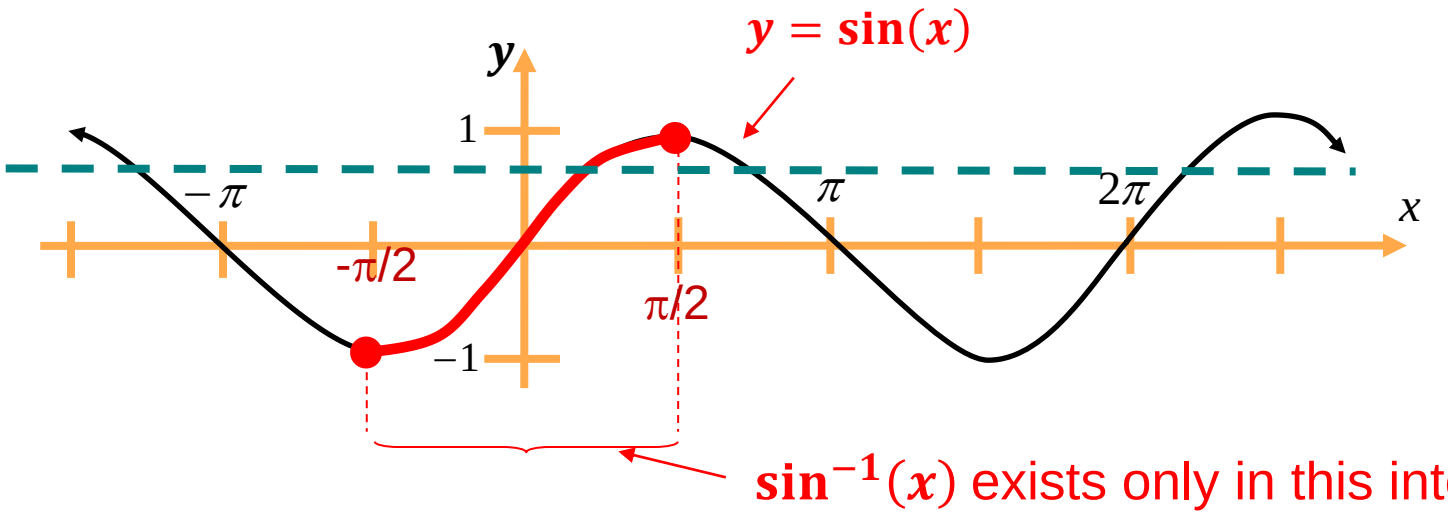
Inverse of $\sin(x)$

For a function to have an inverse, it must be one-to-one

$y = \sin(x)$ $x \in \Re$, **is not one-to-one**

Domain restriction makes $\sin(x)$ one-to-one

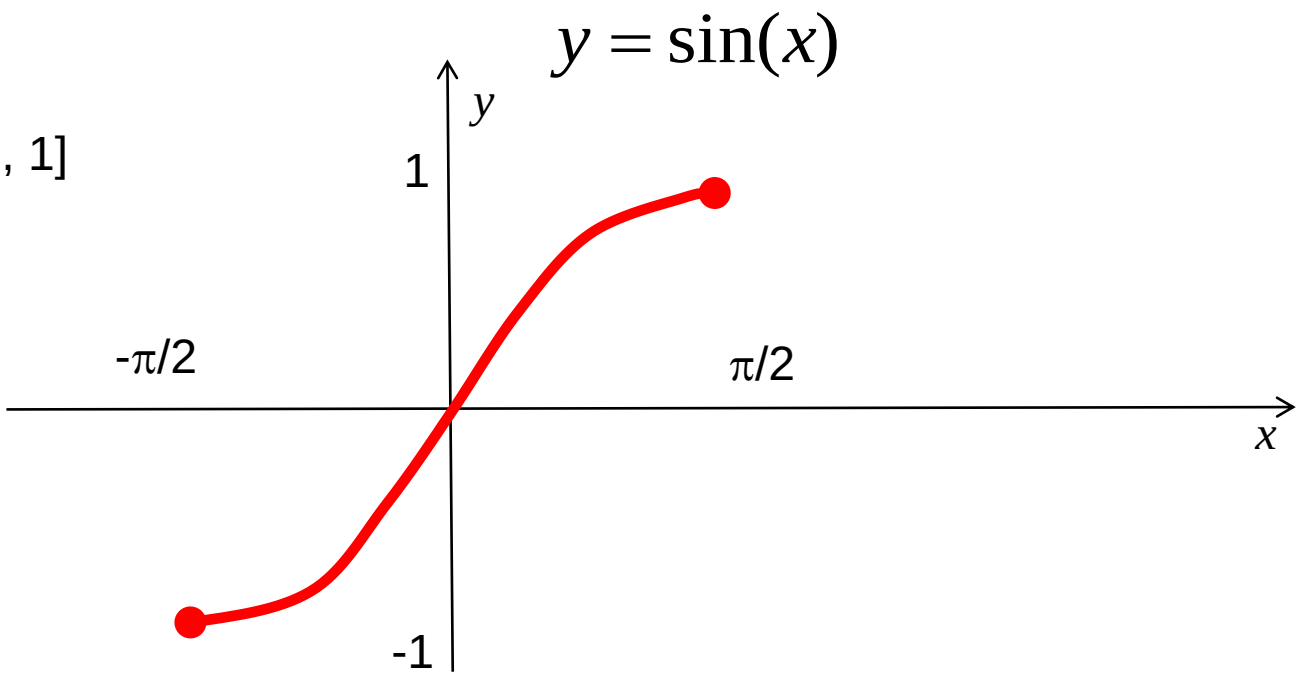
$y = \sin(x)$ $x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ **is one-to-one**



$\sin(x)$: Domain Restrcition

Domain = [–π/2 , π/2]

Range = [–1, 1]

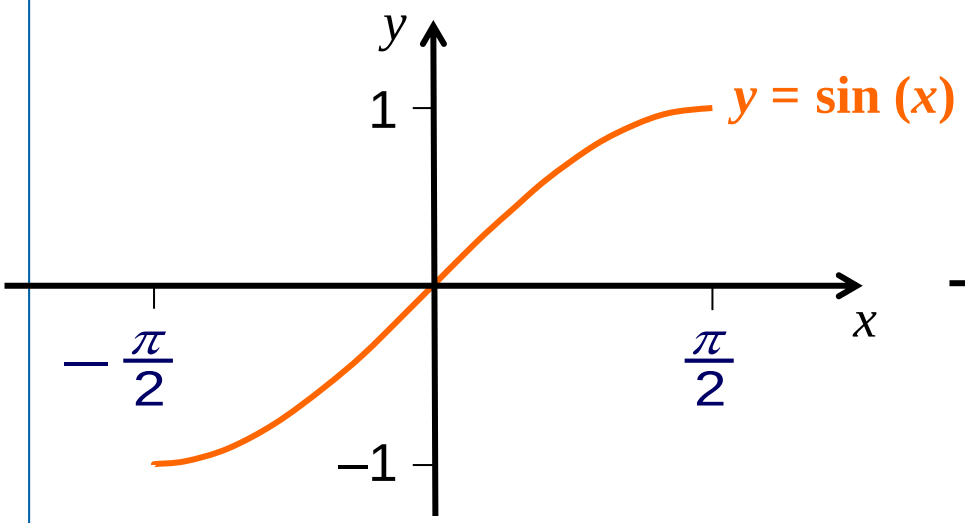


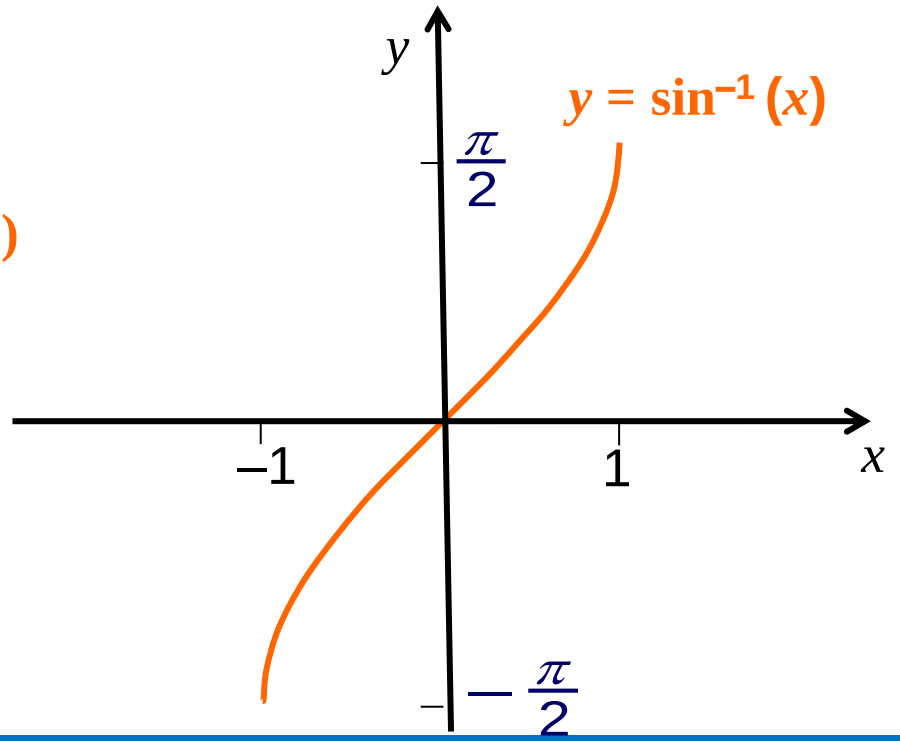
$\sin^{-1}(x)$ exists only in this interval

Inverse of $\sin(x)$: Graphing

$D = \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ and $R = [-1, 1]$

$D = [-1, 1]$ and $R = \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$



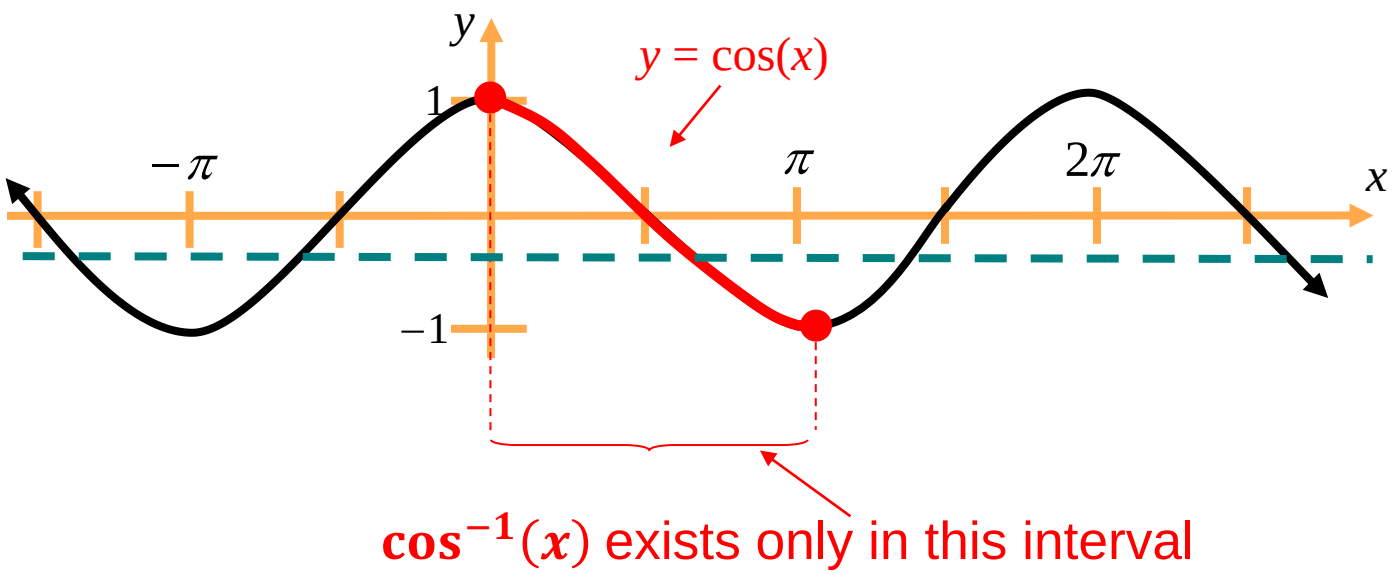


Inverse of $\cos(x)$

$y = \cos(x)$ $x \in \Re$, is **not** one-to-one

Domain restriction makes $\cos(x)$ one-to-one

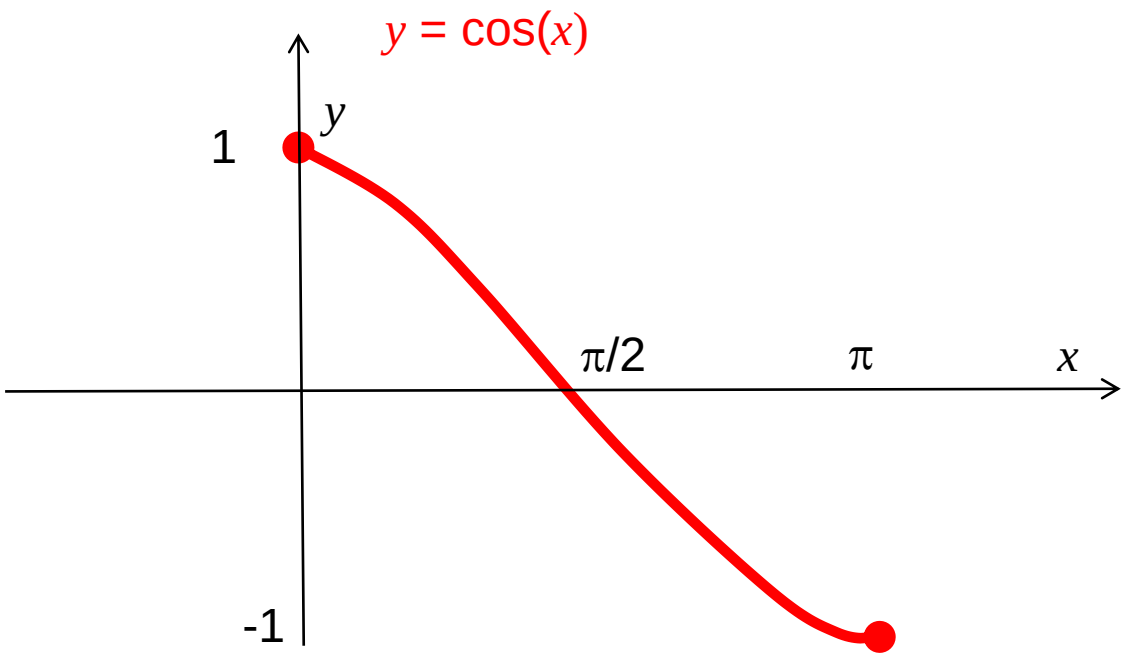
$y = \cos(x)$ $x \in [0, \pi]$ is one-to-one



$\cos(x)$: Domain Restriction

Domain= [0 , π]

Range= [–1, 1]

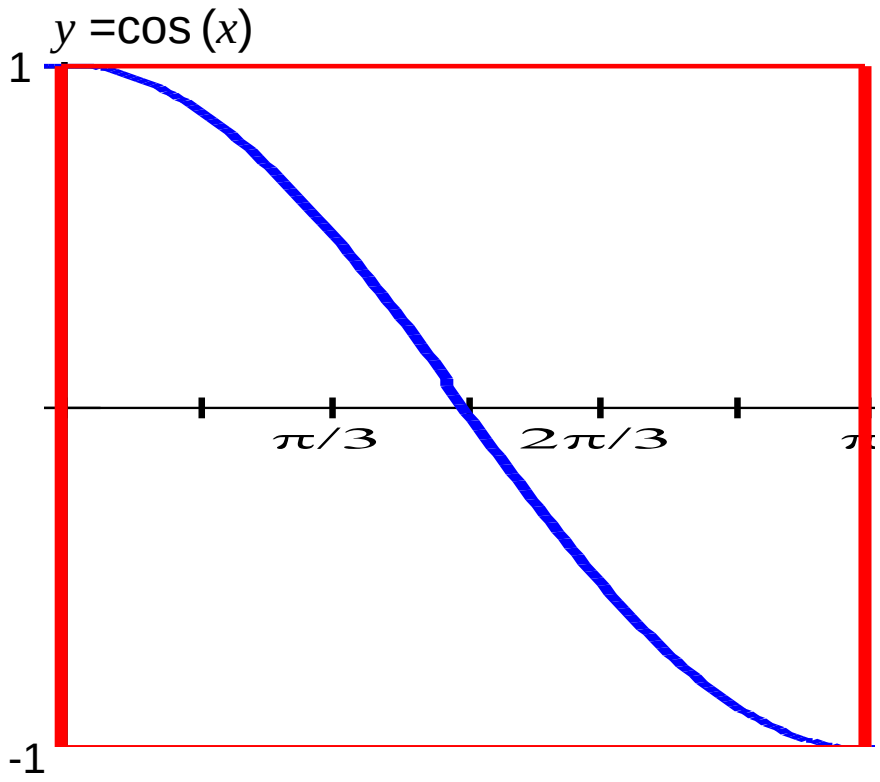


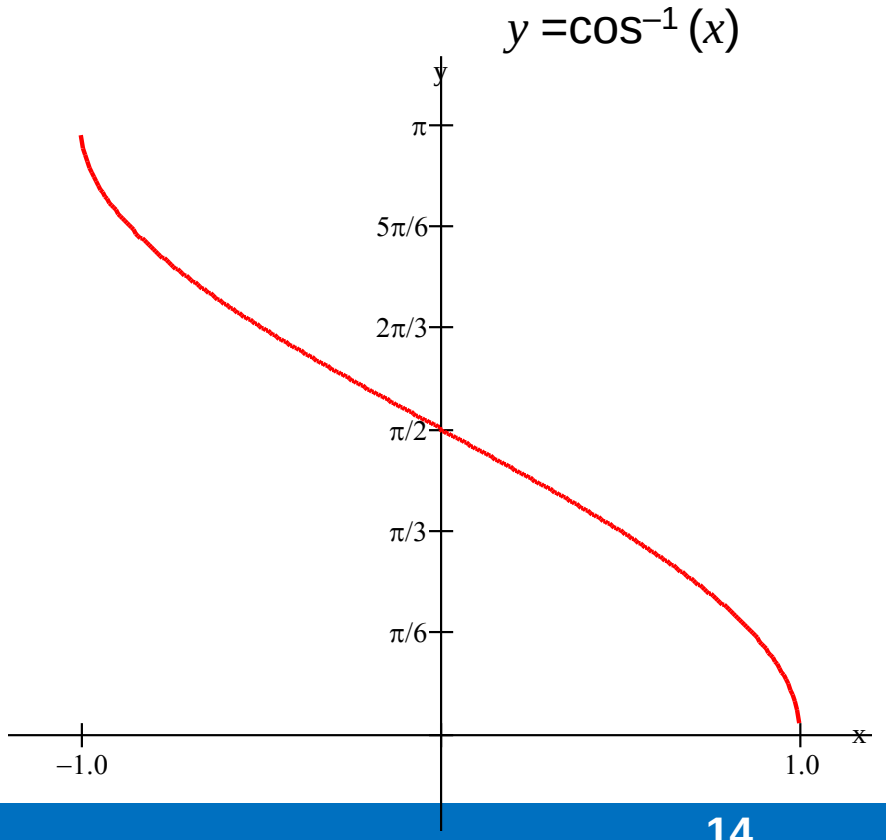
$\cos^{-1}(x)$ exists only in this interval

Inverse of $\cos(x)$: Sketching

$D = [0, \pi]$ and $R = [-1, 1]$

$D = [-1, 1]$ and $R = [0, \pi]$





Inverse Trigonometric Functions: Example

Example **Find the domain of the following function**

$$f(x) = \cos^{-1}(2x - 3)$$

$$-1 \le (2x - 3) \le 1$$ Add 3

$$-1+3 \le (2x - 3)+3 \le 1+3$$ Divide by 2

$$2 \le 2x \le 4$$

$$1 \le x \le 2$$

$$D(f) = [1, 2]$$

Inverse Trigonometric Functions: Example

Example **Find the domain of the following function**

$$f(x) = \frac{\sqrt{x}}{\sin^{-1}(x) - \frac{\pi}{2}}$$

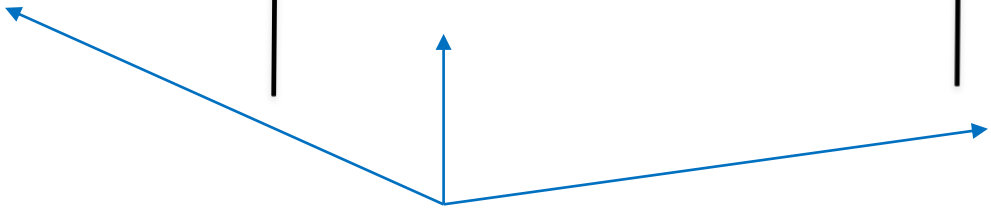
$x \geq 0$

$-1 \leq x \leq 1$

$\sin^{-1}(x) \neq \frac{\pi}{2}$

$x \neq \sin(\frac{\pi}{2})$

$x \neq 1$



Take the intersection

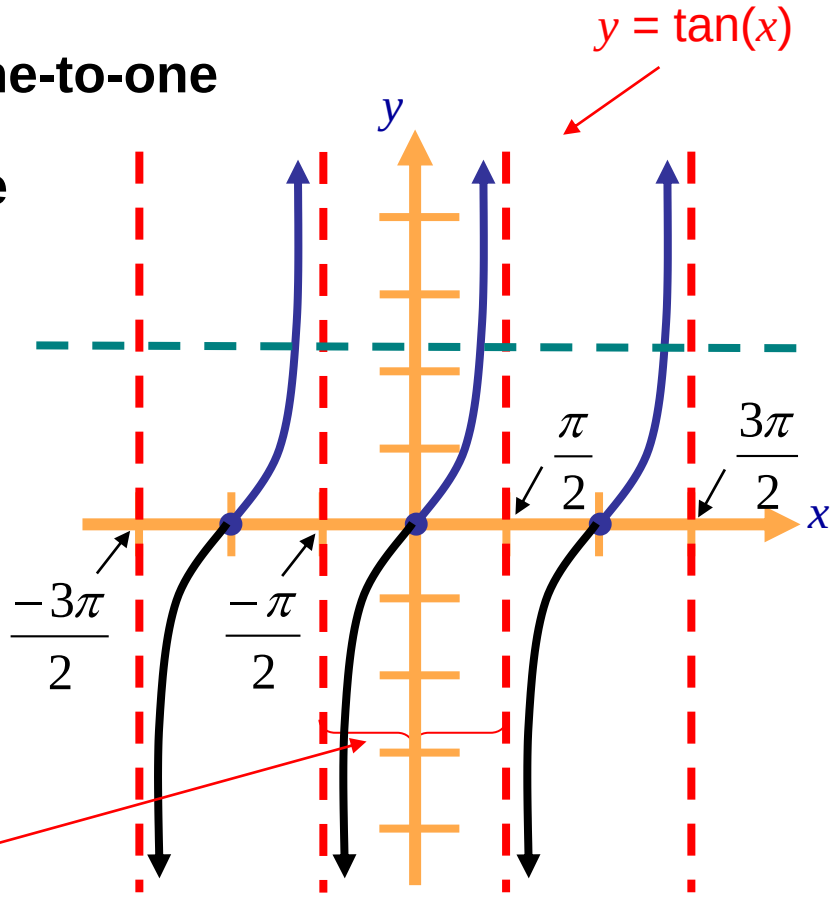
$$D(f) = [0,1)$$

Inverse of $\tan(x)$

$y = \tan(x)$ $x \in \Re$, **is not one-to-one**

Domain restriction makes $\tan(x)$ one-to-one

$y = \tan(x)$ $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$ **is one-to-one**

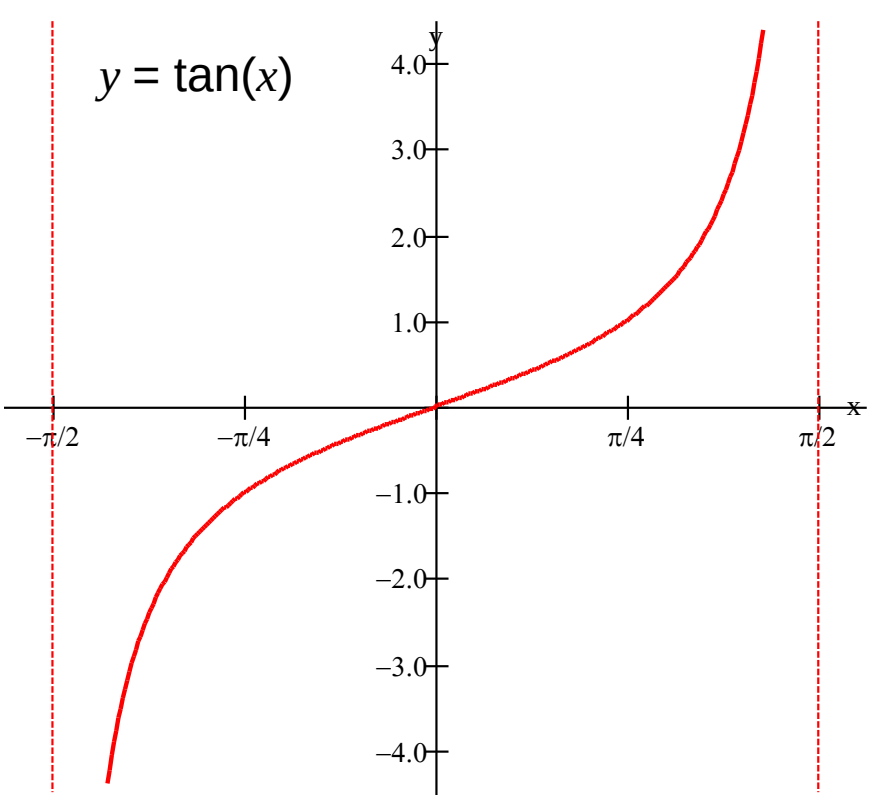


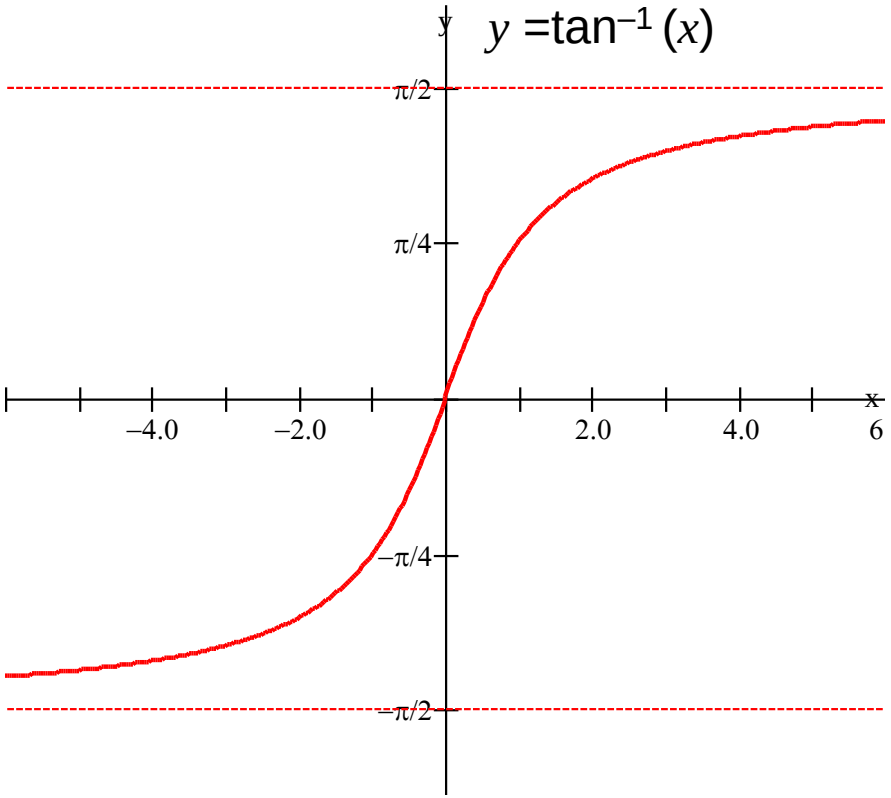
$\tan^{-1}(x)$ exists only in this interval

Inverse of $\tan(x)$: Domain and Range, Graphing

$D = \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ and $R = (-\infty, \infty)$

$D = (-\infty, \infty)$ and $R = \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$





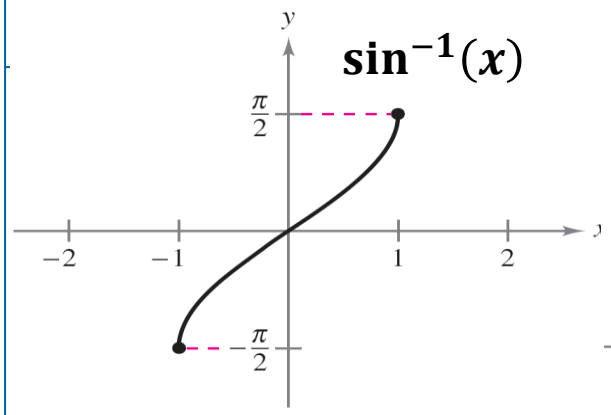
Inverse Trigonometric Functions: Example

Example **Sketch in steps**

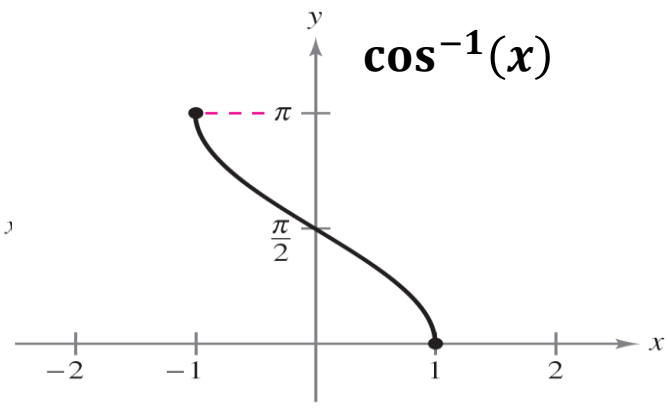
$$y = \frac{1}{\left|\tan^{-1} x\right|}$$

Try to solve!

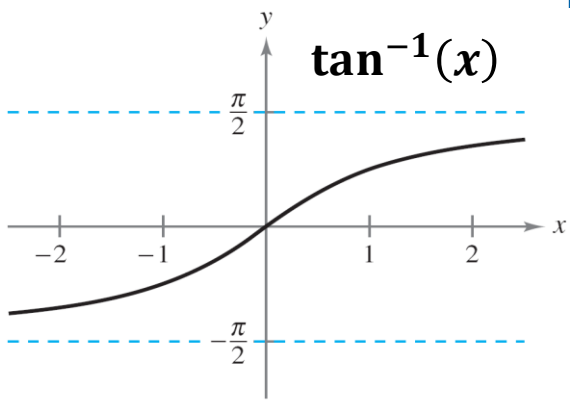
Inverse Trigonometric Functions



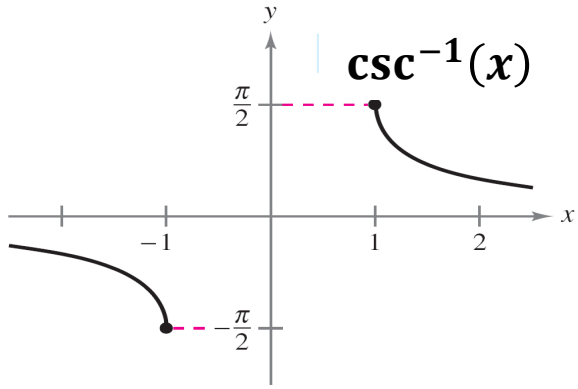
Domain: $[-1, 1]$
Range: $[-\pi/2, \pi/2]$



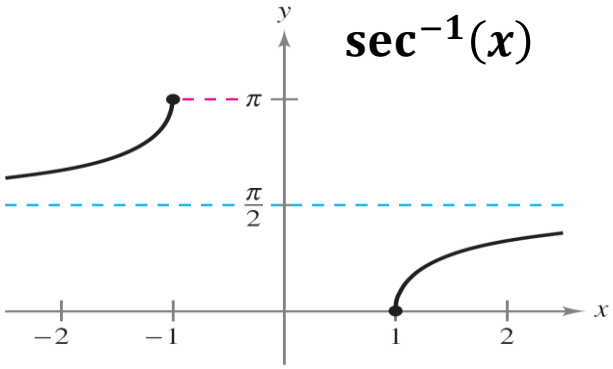
Domain: $[-1, 1]$
Range: $[0, \pi]$



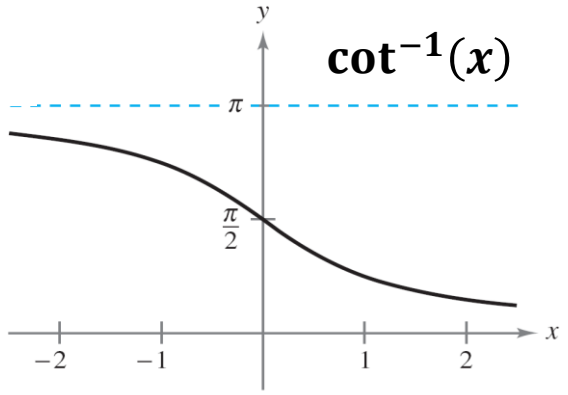
Domain: $(-\infty, \infty)$
Range: $(-\pi/2, \pi/2)$



Domain: $(-\infty, -1] \cup [1, \infty)$
Range: $[-\pi/2, 0) \cup (0, \pi/2]$



Domain: $(-\infty, -1] \cup [1, \infty)$
Range: $[0, \pi/2) \cup (\pi/2, \pi]$



Domain: $(-\infty, \infty)$
Range: $(0, \pi)$

Inverse Trigonometric Functions: Symmetry

$\sin^{-1}(-x) = -\sin^{-1}(x)$ Odd function

$\tan^{-1}(-x) = -\tan^{-1}(x)$ Odd function

$\csc^{-1}(-x) = -\csc^{-1}(x)$ Odd function

Other functions are neither even nor odd.

Note

The inverse functions can never be even.

Inverse Trigonometric Functions
Cancellation Equations for the Inverse Function

$\sin(\sin^{-1} x) = x \qquad -1 \le x \le 1$

$\sin^{-1}(\sin x) = x \qquad -\frac{\pi}{2} < x < \frac{\pi}{2}$

$\cos(\cos^{-1} x) = x \qquad -1 \le x \le 1$

$\cos^{-1}(\cos x) = x \qquad 0 \le x \le \pi$

$\tan(\tan^{-1} x) = x \qquad -\infty < x < \infty$

$\tan^{-1}(\tan x) = x \qquad -\frac{\pi}{2} < x < \frac{\pi}{2}$

$\sin(\sin^{-1}(0.5)) = 0.5$

$\sin(\sin^{-1}(2)) \neq 2$

$\sin^{-1}(\sin(\frac{\pi}{3})) = \frac{\pi}{3}$

$\sin^{-1}(\sin(\pi)) \neq \pi$

$\tan^{-1}(\tan(\pi)) \neq \pi$

$\tan^{-1}(\tan(\frac{\pi}{6})) = \frac{\pi}{6}$

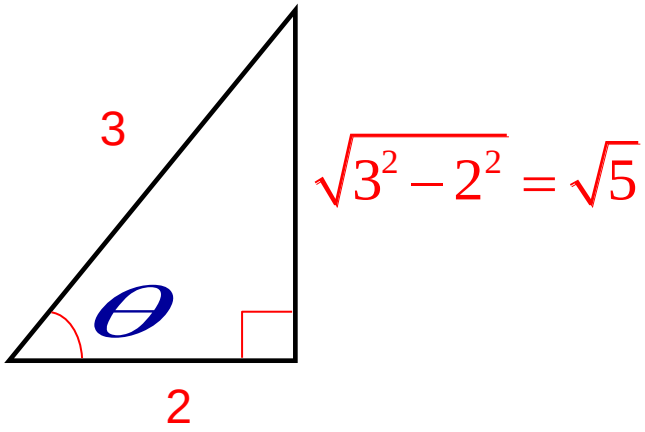
$\tan(\tan^{-1}(2\pi)) = 2\pi$

Inverse Trigonometric Functions: Example

Example Find the value of $\tan\left(\cos^{-1}\left(\frac{2}{3}\right)\right)$

$$\text{Let } \theta = \cos^{-1}\frac{2}{3}, \text{ then } \cos\theta = \frac{2}{3} = \frac{adj}{hyp}$$

$$\tan\left(\cos^{-1}\left(\frac{2}{3}\right)\right) = \tan\theta = \frac{opp}{adj} = \frac{\sqrt{5}}{2}$$

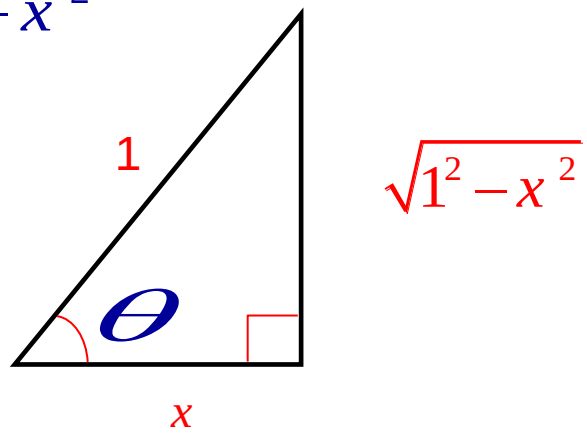


Inverse Trigonometric Functions: Example

Example **Show that** $\sin\left(\cos^{-1}(x)\right)=\sqrt{1-x^2}$

$$\text{Let } \theta = \cos^{-1}\text{x}, \text{ then } \cos\theta = x = \frac{adj}{hyp}$$

$$\sin\left(\cos^{-1}x\right) = \sin\theta = \frac{opp}{hyp} = \frac{\sqrt{1-x^2}}{1} = \sqrt{1-x^2}$$



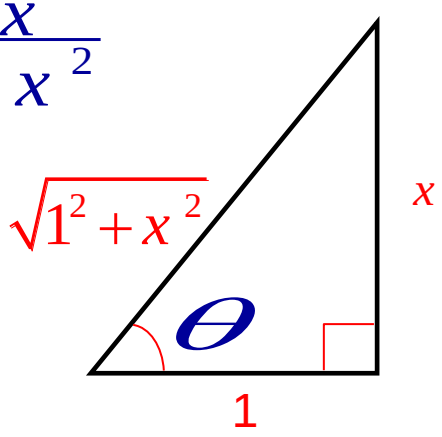
Inverse Trigonometric Functions: Example

Example **Show that** $\sin\left(2\tan^{-1}(x)\right)=\frac{2x}{1+x^2}$

Let $\theta = \tan^{-1} x$, then $\tan\theta = x$

$$\sin\left(2\tan^{-1}x\right) = \sin 2\theta = 2\sin(\theta)\cos(\theta)$$

$$\sin\left(2\tan^{-1}x\right) = 2\frac{x}{\sqrt{1+x^2}}\frac{1}{\sqrt{1+x^2}} = \frac{2x}{1+x^2}$$



Inverse Trigonometric Functions Identities

Prove that

$$\sin^{-1} x + \cos^{-1} x = \pi/2$$

$$\tan^{-1} x + \cot^{-1} x = \pi/2$$

$$\sec^{-1} x + \csc^{-1} x = \pi/2$$

Try to solve!

$$\csc^{-1} x = \sin^{-1}(1/x)$$

$$\sec^{-1} x = \cos^{-1}(1/x)$$

$$\cot^{-1} x = \tan^{-1}(1/x) \ , \ x > 0$$

Inverse Trigonometric Functions Identities

proof $\sec^{-1} x = \cos^{-1}\left(\frac{1}{x}\right)$

$\mathbf{let}\ \ y = \sec^{-1} x$

$\sec y = x$

$\cos y = \frac{1}{x}$

$y = \cos^{-1}(1/x)$

Derivatives of Inverse Trigonometric Functions

$$\frac{d}{dx}\sin^{-1}x = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}\cos^{-1}x = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}\tan^{-1}x = \frac{1}{1+x^2}$$

$$\frac{d}{dx}\cot^{-1}x = -\frac{1}{1+x^2}$$

$$\frac{d}{dx}\sec^{-1}x = \frac{1}{|x|\sqrt{x^2-1}}$$

$$\frac{d}{dx}\csc^{-1}x = -\frac{1}{|x|\sqrt{x^2-1}}$$

Proof

- *We want to find the derivative of* $y = \sin^{-1} x.$

- *We will rewrite this as* $\sin y = x$ *and take the derivative.*

$$\cos y \cdot y' = 1 \Rightarrow \quad y' = \frac{1}{\cos y} \Rightarrow \quad y' = \frac{1}{\sqrt{1 - \sin^2 y}}$$

$$y' = \frac{1}{\sqrt{1 - x^2}}$$

$$\frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1 - x^2}}$$

Similarly

$$\frac{d}{dx} \cos^{-1} x = -\frac{1}{\sqrt{1 - x^2}}$$

Finding Derivatives of Inverse Trigonometric Functions

Example **Find the derivative of** $y = \cos^{-1} x$

$$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$$

$$[\sin^{-1} x]' + [\cos^{-1} x]' = 0$$

$$[\cos^{-1} x]' = \frac{-1}{\sqrt{1-x^2}}$$

Proofs: $y = \tan^{-1} x$

$y = \tan^{-1} x \quad \Rightarrow \tan y = x$

$\Rightarrow \quad (\sec^2 y)y' = 1$

$$y' = \frac{1}{\sec^2 y} = \frac{1}{1+\tan^2 y} = \frac{1}{1+x^2}$$

$$[\tan^{-1} x]' = \frac{1}{1+x^2}$$

Similarly

$$\frac{d}{dx}\cot^{-1} x = -\frac{1}{1+x^2}$$

Finding Derivatives of Inverse Trigonometric Functions

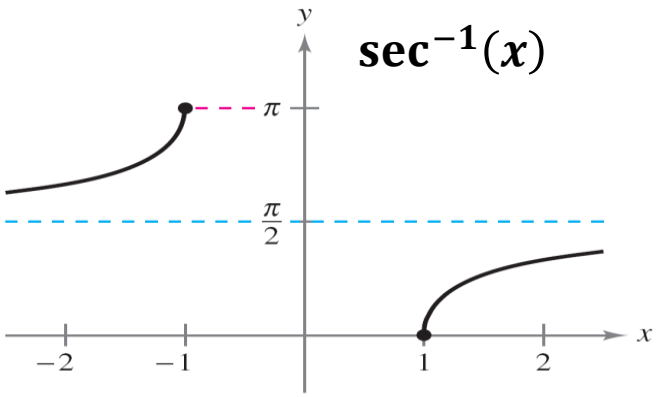
Example **Find the derivative of** $y = \sec^{-1} x$

$$\sec y = x$$

$$(\sec y \tan y)y' = 1$$

$$y' = \frac{1}{\sec y \tan y} = \frac{1}{\sec y \sqrt{\sec^2 y - 1}} = \frac{1}{x\sqrt{x^2-1}}$$

$$[\sec^{-1} x]' = \frac{1}{|x|\sqrt{x^2-1}}$$



Domain: $(-\infty, -1] \cup [1, \infty)$
Range: $[0, \pi/2) \cup (\pi/2, \pi]$

Derivatives of Inverse Trigonometric Functions Examples

Example **Find** y' **if** $y = x\tan^{-1}x + \sqrt{\sin^{-1}x}$

$$y' = (1)\tan^{-1}x + \frac{x}{1+x^2} + (0.5)(\sin^{-1}x)^{-0.5}\frac{1}{\sqrt{1-x^2}}$$

Example **Find** y' **if** $y = e^{\sec^{-1}x} + \ln(\tan^{-1}x)$

$$y' = e^{\sec^{-1}x}\frac{1}{|x|\sqrt{x^2-1}} + \frac{1}{\tan^{-1}x}\frac{1}{1+x^2}$$

Derivatives of Inverse Trigonometric Functions Examples

Example **Find** y' **if** $y = \cos^{-1} x + \cos(x^{-1}) + (\cos x)^{-1}$

$$y' = -\frac{1}{\sqrt{1-x^2}} - \sin(x^{-1})(-x^{-2}) + \sec x \tan x$$

[**note** $(\cos x)^{-1} = \sec x$]

Derivatives of Inverse Trigonometric Functions

- *With the chain rule these formulas become:*

$$\frac{d}{dx}\sin^{-1}u = \frac{1}{\sqrt{1-u^2}}\frac{du}{dx}$$

$$\frac{d}{dx}\tan^{-1}u = \frac{1}{1+u^2}\frac{du}{dx}$$

$$\frac{d}{dx}\sec^{-1}u = \frac{1}{|u|\sqrt{u^2-1}}\frac{du}{dx}$$

$$\frac{d}{dx}\cos^{-1}u = -\frac{1}{\sqrt{1-u^2}}\frac{du}{dx}$$

$$\frac{d}{dx}\cot^{-1}u = -\frac{1}{1+u^2}\frac{du}{dx}$$

$$\frac{d}{dx}\csc^{-1}u = -\frac{1}{|u|\sqrt{u^2-1}}\frac{du}{dx}$$

Derivatives of Inverse Trigonometric Functions Examples

Example **Find** y' **if** $y = \sin^{-1}(e^x) - \sec^{-1}(x^2 + 1)$

$$y' = \frac{1}{\sqrt{1-(e^x)^2}} e^x - \frac{1}{x^2+1\sqrt{(x^2+1)^2-1}} * 2x$$

Example **Find** y' **if** $y = \tan^{-1}\sqrt{x} + \cot^{-1}\left(\frac{1}{x}\right)$

$$y' = \frac{1}{1+\left(\sqrt{x}\right)^2} \frac{1}{2\sqrt{x}} - \frac{1}{1+\left(\frac{1}{x}\right)^2} \left(-\frac{1}{x^2}\right)$$

Derivatives of Inverse Trigonometric Functions Examples

Example Find y' if $y - x = \cot^{-1}\frac{y}{x}$

$$y' - 1 = \frac{-1}{1 + \left(\frac{y}{x}\right)^2}\left(\frac{xy' - y}{x^2}\right) = \frac{y - xy'}{x^2 + y^2}$$

$$\left(1 + \frac{x}{x^2+y^2}\right)y' = \frac{y}{x^2+y^2} + 1$$

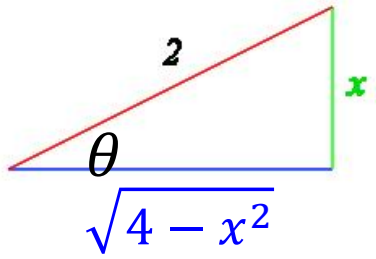
$$y' = \frac{x^2+y^2+y}{x^2+y^2+x}$$

Derivatives of Inverse Trigonometric Functions
Examples

Example **Find** y' **and** y'' **if** $y = \sin\left(\cot^{-1}\frac{\sqrt{4-x^2}}{x}\right)$

Let $\theta = \cot^{-1}\frac{\sqrt{4-x^2}}{x}$

i.e. $\cot\theta = \frac{\sqrt{4-x^2}}{x}$



$y = \sin\theta = \frac{x}{2} \quad \Rightarrow \quad y' = \frac{1}{2} \quad \Rightarrow \quad y'' = 0$

See you next week



